

Hardware Components per Module

what the RTL instantiates — memories, registers, multipliers, muxes, counters, FSMs (5-qubit build)

CONTROL PLANE

qml_accelerator (top / sequencer + PSR)		
Instruction ROM	imem	64-bit × 256
Parameter RAM	pmem	16-bit × 64
Program counter	pc	8-bit counter
Instruction latch	cur_*	op/qa/qb/mask/pidx
Result registers		meas_reg, E+, E-, result — 32-bit ×4
Override adders (±π/2)		3 adders + 3 compare-muxes
Gradient path		subtractor + »1 shift
Sequencer FSM	rst_s	8 states
Param-shift FSM	pst	6 states

drives

DATAPATH

gate_gen (opcode → matrix)		
Opcode latch	op_r	8-bit
Angle regs	ang[0:3]	4 × 16-bit
cos/sin store	cc,ss	8 × 16-bit
Coefficient qmul		1 multiplier (U2/U3)
Fixed-gate ROM		constants (X,H,S,T,SWAP...)
Matrix-select mux		8 fields (2×2) + 32 (4×4)
Call counter	k	3-bit
FSM	fst	4 states

instantiates

cordic (cos/sin engine)		
Pipeline x,y,z	xr,yr,zr	17 × 32-bit each
Flags	fr,vr	17 × 1-bit each
arctan ROM		16 × 32-bit constants
Per-stage add/sub		3 × 16 stages
Per-stage shifters		2 × 16 stages
Sign muxes		16 (one per stage)
Arg-reduce + round/sat		3 add/sub + rounding
no multipliers — shift/add only		

qpu_core (state-vector engine)		
State memory	sr, si	16-bit × 32, ×2 banks
Sweep counter	cnt	6-bit
Accumulator	acc	48-bit
Expectation out	exp_q28	32-bit
MAC cells		6 × cmac (see below)
Measurement amp ²		2 multipliers + adder
Parity tree		XOR reduction
Read / write-back muxes		1Q pair / 2Q quad select
FSM	st	6 states

instantiates ×6

cmac (×6 instances)		
Signed multipliers		16 × (16×16-bit)
Adder tree		real + imag accumulate
Round + saturate		Q2.28 → Q1.14
u_bf0,u_bf1 (2×2)		u_r0...u_r3 (4×4)

TOTAL (5-qubit build)

Multipliers (16×16)	≈ 98 (16×6 MAC + 2)
Adders / subtractors	≈ 100
Flip-flops	≈ 2,300 (CORDIC ≈ 1,700)
Memory	≈ 18.4 kbit
MAC cells · FSMs · counters	6 · 4 · 3
scales with qubits (2 ⁿ)	only the memories